From the Binomial Theorem to Coq: An Educational Explanation of Version 15.3 of the Conditional Model of Fermat's Last Theorem

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Abstract

This document presents an educational exposition of version 15.3 of a conditional reconstructive model surrounding Fermat's Last Theorem. The text is written for a strong math lyceum graduate and explains not the standard Wiles–Taylor proof, but a separate scheme: Fermat's equation, parameterization via m, p, the odd binomial core $S_n(m, p)$, two natural real normalizations of the same core, the exact relation

$$l^n = q^n \frac{2^{n-1}}{n},$$

the compatibility equivalence of version 15.3

$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1},$$

and the machine-verifiable final chain in Coq

$$n = 2^{n-1} \quad \Longrightarrow \quad n \in \{1, 2\}.$$

The real, integer, and modular levels are considered separately to avoid confusing equalities over $\mathbb{R}$ and $\mathbb{Z}$ with modular congruences. The main goal of this text is not to declare an unconditional new proof of Fermat's Last Theorem, but to clearly show the conditional logical structure: given two natural normalizations of the same odd binomial core, the equality of residual scales is compatible with the structure of binomial coefficients if and only if $n = 2^{n-1}$, and is therefore possible only in the linear and quadratic cases.

Keywords: Fermat's Last Theorem; conditional mathematical model; binomial theorem; odd binomial core; real factorization; two normalizations of the core; equality of residual scales; coefficient symmetry compatibility obstacle; $l^n = q^n \Longleftrightarrow n = 2^{n-1}$; Coq; formal verification; modular remark; integer specialization; educational exposition for a math lyceum.

Contents

Preliminary Honest Remark on the Status of the Model

Fermat's Last Theorem has already been proven by modern mathematics in the works of Andrew Wiles and Richard Taylor. The modern proof uses deep methods of modular forms and elliptic curves. This educational text does not replace that proof and is not an unconditional new proof of the theorem.

A different task is considered here: to carefully present a conditional reconstructive model in which the odd binomial core is isolated and two natural real normalizations of this core are compared. Therefore, the key difference must be established immediately:

The unconditional proof asserts FLT without additional hypotheses. **The conditional model of version 15.3** shows: given two real normalizations of the same odd binomial core, the equality of residual scales $l^n = q^n$ is compatible with the binomial structure if and only if $n = 2^{n-1}$; therefore, under such a compatibility requirement, an exponent $n > 2$ is impossible.

There are two parts to this model:

1. **Algebraic-reconstructive part**: from the symmetric notation of the difference of powers, the odd binomial core $S_n(m, p)$ and its two normalizations are extracted.
2. **Formally verifiable part**: after accepting the two factorizations, Coq verifies the exact relation between the residual scales, the equivalence

$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1},$$

and then the final conclusion $n \in \{1, 2\}$.

This separation is important. Computer verification does not replace the mathematical justification that a hypothetical positive integer solution is obliged to preserve the equality of residual scales $l^n = q^n$, but it guarantees that further logical transitions within the chosen conditional scheme are executed strictly.

The Main Idea of Version 15.3

The current version is built around a more precise central scheme:

$$\text{core} = n\, l^n, \qquad \text{core} = 2^{n-1} q^n \quad \Longrightarrow \quad l^n = q^n \frac{2^{n-1}}{n}.$$

From here, the main compatibility criterion is obtained:

$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1}.$$

And then an elementary comparison of growth gives

$$n = 2^{n-1} \quad \Longrightarrow \quad n \in \{1, 2\}.$$

The main points are as follows:

1. The main object remains the odd binomial core $S_n(m, p)$.
2. Two residual real scales are used: l and q.

3. The first normalization isolates the linear binomial coefficient:

$$S_n(m,p) = n\,l^n.$$

4. The second normalization isolates the total mass of odd binomial coefficients:

$$S_n(m,p) = 2^{n-1}q^n.$$

5. Comparing the two normalizations gives an exact relation:

$$l^n = q^n\frac{2^{n-1}}{n}.$$

6. The central point of version 15.3 is formulated as the coefficient symmetry compatibility obstacle:

$$l^n = q^n \quad\Longleftrightarrow\quad n = 2^{n-1}.$$

7. The equality of the two residual scales is compatible only with $n = 1$ and $n = 2$.
8. The modular level is presented separately. The real scale q is not a module; for the module, the notation modq is used.
9. The Coq description is provided in terms of the `GlobalNormalization.v` file of version 15.3: `residual_scale_equality`, `coefficient_mass_equality`, `coefficient_symmetry_compatibility_iff`, `CoefficientSymmetryData` and related lemmas.

For a math lyceum graduate, this text is useful as an example of mathematical culture: a strong idea must be written in such a way that the definitions, the scope of applicability, conditional spots, boundaries of the model, and machine-verifiable consequences are all visible.

1 Historical Statement of the Problem

Fermat's Last Theorem states that the equation

$$x^n + y^n = z^n \tag{1}$$

has no solutions in positive integers x, y, z for integer $n > 2$.

For $n = 1$, solutions are obvious: for example, $1 + 2 = 3$. For $n = 2$, Pythagorean triples exist:

$$3^2 + 4^2 = 5^2.$$

But starting with cubes, the situation changes: one cannot represent a cube as the sum of two cubes, a fourth power as the sum of two fourth powers, and so on.

Fermat formulated this statement in the margins of his copy of Diophantus's *Arithmetica* and wrote that he had found a "truly marvelous demonstration," but the margins of the book were too narrow to contain it. From a historical perspective, it is natural to ask: could Fermat have had an elementary idea that he did not write down in modern form?

This document does not claim that Fermat's exact historical proof has been found. It offers an educational reconstruction of a conditional scheme showing how a strict implication might arise from the binomial theorem, the symmetry of two terms, and the comparison of two real scales.

2 How a Math Lyceum Graduate Should Read This Document

To understand the text, it is sufficient to confidently master the following topics:

- powers, roots, and natural numbers;
- the binomial theorem;
- elementary divisibility;
- comparing the growth of functions;
- a basic understanding of what a formal proof verification is.

The main methodological principle: we will distinguish three things.

1. **Algebraic identity**: an equality obtained by expanding brackets or changing variables.
2. **Condition of the model**: an additional statement that must be accepted or proven separately.
3. **Formal conclusion**: a consequence that Coq verifies after all prerequisites are explicitly written down.

This distinction protects against a typical mistake in amateur proofs: a beautiful observation cannot be imperceptibly turned into a proven theorem. In this version, the central mathematical nexus is named directly: it is the status of the equality of residual scales $l^n = q^n$ and its precise equivalence to $n = 2^{n-1}$ under two normalizations of the same core.

3 Mapping Between the Article, the Educational Text, and Coq

To avoid getting lost in the notation, it is useful to keep the following map in front of you.

Level	**Mathematical Meaning**	**Name in Coq**
Real parameterization	If $z = m^n + p^n$, $x = m^n - p^n$, then $z + x = 2m^n$, $z - x = 2p^n$ over $\mathbb{R}$.	`sum_diff_from_parameters_R`
Integer specialization	The same formula over $\mathbb{Z}$; for integer m, p, parity conditions arise.	`sum_diff_from_parameters_Z`; `parity_condition_Z`
Verifying the impossibility of integer parameters	If $z + x$ or $z - x$ has incorrect parity, integer m, p do not exist in this form.	`no_parameters_if_parity_violation`; `no_parameters_if_odd`; `no_parameters_for_example`
First normalization of the core	The odd core is written as core $= n\, l^n$ over $\mathbb{R}$.	`real_core_factorization`
Second normalization of the core	The same core is written as core $= 2^{n-1} q^n$ over $\mathbb{R}$.	`coefficient_mass_factorization`
Comparison of the two normalizations	From the two notations of one core it follows $$l^n = q^n \frac{2^{n-1}}{n}.$$	`two_scale_factorizations_relation`
Equality of residual scales	The equality $l^n = q^n$.	`residual_scale_equality`

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Level	**Mathematical Meaning**	**Name in Coq**
Equality of coefficient masses	The equality $n = 2^{n-1}$.	`coefficient_mass_equality`
Forward direction of compatibility	Two factorizations, $l^n = q^n$ and $q > 0$ entail $n = 2^{n-1}$.	`residual_scale_equality_forces_coefficient_mass_equality`
Reverse direction of compatibility	Two factorizations and $n = 2^{n-1}$ entail $l^n = q^n$.	`coefficient_mass_equality_forces_residual_scale_equality`
Central equivalence of version 15.3	Under two factorizations and $q > 0$, we have $$l^n = q^n \iff n = 2^{n-1}.$$	`coefficient_symmetry_compatibility_iff`
Packaging model data	In a single structure are stored n, the core, l, q, positivity, both factorizations, and the equality of residual scales.	`CoefficientSymmetryData`
Final growth	From $n = 2^{n-1}$ follows $n = 1$ or $n = 2$.	`binary_scaling_roots_only_one_two`
Exclusion of large exponents	From compatibility data and $n > 2$, a contradiction follows.	`coefficient_symmetry_data_excludes_high_exponent`; `coefficient_symmetry_compatibility_excludes_high_exponent`
Modular caveat (Remark)	An integer equality entails a modular congruence, but the reverse is false.	`ModularRemark`; `modular_congruence_not_integer_equality`

The educational text explains the idea in human language. The article fixes the mathematical formulation. The Coq file checks only the implications that are explicitly written in the code.

4 Parameterization via m and p

Suppose hypothetically there is a positive integer solution to Fermat's equation for some $n > 2$:

$$x^n + y^n = z^n.$$

Let us rewrite it as

$$z^n - x^n = y^n. \tag{2}$$

Now let us introduce parameters m and p using the system

$$\begin{cases} m^n + p^n = z, \\ m^n - p^n = x. \end{cases} \tag{3}$$

Adding and subtracting these equations:

$$2m^n = z + x, \qquad 2p^n = z - x.$$

Formally, we obtain

$$m = \sqrt[n]{\frac{z+x}{2}}, \qquad p = \sqrt[n]{\frac{z-x}{2}}. \tag{4}$$

Remark 4.1. If m and p are allowed as real numbers, such a reconstruction is an algebraic notation after choosing a suitable branch of the root. If we additionally require $m, p \in \mathbb{Z}$, conditions arise: $z + x$ and $z - x$ must be even, and the numbers $(z + x)/2$ and $(z - x)/2$ must be exact n-th powers in $\mathbb{Z}$.

Substituting (3) into (2), we get

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n. \tag{5}$$

This is exactly where the binomial part begins.

5 Three Levels of Numbers: $\mathbb{R}$, $\mathbb{Z}$, and the Modular Level

The same expression can live in different mathematical worlds. In this model, it is important to distinguish three levels.

5.1 Real Level

Here $m, p, l, q \in \mathbb{R}$. We can extract roots, introduce real scales, and write down factorizations of the form

$$S_n(m,p) = n\, l^n, \qquad S_n(m,p) = 2^{n-1} q^n.$$

At this level, the notation means the equality of real values.

5.2 Integer Level

Here conditions $x, y, z \in \mathbb{N}$ are additionally discussed, and if required, $m, p \in \mathbb{Z}$. For example, from

$$z = m^n + p^n, \qquad x = m^n - p^n$$

with $m, p \in \mathbb{Z}$ it follows

$$z + x = 2m^n, \qquad z - x = 2p^n.$$

Therefore, $z + x$ and $z - x$ must be even. If this is not the case, integer parameters m, p in this form cannot exist.

Example 5.1. Let $n = 3$, $z = 2$, $x = 1$. Then

$$\frac{z+x}{2} = \frac{3}{2}, \qquad \frac{z-x}{2} = \frac{1}{2}.$$

Real roots exist, but there are no integer m, p satisfying

$$2 = m^3 + p^3, \qquad 1 = m^3 - p^3.$$

It is exactly this type of difference that the lemma `no_parameters_for_example` fixes.

5.3 Modular Level

At the modular level, residues are considered. To avoid confusing the module with the real scale q, we will denote the module by modq:

$$x^n + y^n \equiv z^n \pmod{\text{modq}}.$$

Such a congruence can be true even when the equality $x^n + y^n = z^n$ in integers is false. Therefore, modular examples by themselves are not counterexamples to FLT.

6 The Binomial Theorem and the Odd Binomial Core

Recall the Binomial theorem formula:

$$(A+B)^n = \sum_{r=0}^{n} \binom{n}{r} A^{n-r} B^r. \tag{6}$$

For the difference we have

$$(A-B)^n = \sum_{r=0}^{n} (-1)^r \binom{n}{r} A^{n-r} B^r. \tag{7}$$

Subtracting (7) from (6), we get

$$(A+B)^n - (A-B)^n = \sum_{r=0}^{n} \bigl(1-(-1)^r\bigr) \binom{n}{r} A^{n-r} B^r \tag{8}$$

$$= 2 \sum_{\substack{1\le r\le n \\ r \text{ odd}}} \binom{n}{r} A^{n-r} B^r. \tag{9}$$

In our case

$$A = m^n, \qquad B = p^n.$$

Therefore

$$y^n = 2 \sum_{\substack{1\le r\le n \\ r \text{ odd}}} \binom{n}{r} (m^n)^{n-r} (p^n)^r. \tag{10}$$

Definition 6.1 (Odd binomial core)**.** The odd binomial core will be called the sum

$$S_n(m,p) = \sum_{\substack{1\le r\le n \\ r \text{ odd}}} \binom{n}{r} (m^n)^{n-r} (p^n)^r. \tag{11}$$

Then

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m,p), \tag{12}$$

and consequently,

$$y^n = 2S_n(m,p).$$

Intuitively, $S_n(m,p)$ is the part of the binomial expansion that survives after subtracting two symmetric powers. Even terms cancel out, odd ones double.

7 The First Real Normalization: $S_n(m,p) = n\, l^n$

For each odd index r, the binomial coefficient can be written as

$$\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1}. \tag{13}$$

This allows us to extract the common factor n from the odd core in real-rational algebra.

Therefore, the first normalization is introduced:

$$S_n(m,p) = n\, l^n. \tag{14}$$

Here $l > 0$ is a real residual scale. It shows what part of the core remains after extracting the linear factor n.

If $n > 0$, then formally

$$l^n = \frac{S_n(m,p)}{n}. \tag{15}$$

For a positive right-hand expression, we can take the positive real branch of the root.

From $y^n = 2S_n(m,p)$ and (14) we obtain

$$y^n = 2n\, l^n. \tag{16}$$

In Coq, this corresponds to the definition `real_core_factorization`. It means:

$$0 < n, \qquad \text{core} = n\, l^n$$

over the real numbers.

8 Why This Is Not Integer Divisibility

The notation

$$S_n(m,p) = n\, l^n$$

over $\mathbb{R}$ does not automatically mean

$$n \mid S_n(m,p)$$

in $\mathbb{Z}$.

This is a fundamentally important distinction. The first statement is a real definition of the scale l. The second statement is integer divisibility, which requires a separate proof and might generally be false.

Example 8.1 (Check for $n = 6$)**.** For $n = 6$, $m = 1$, $p = 1$, the odd sum of binomial coefficients is

$$\binom{6}{1} + \binom{6}{3} + \binom{6}{5} = 6 + 20 + 6 = 32.$$

The number 32 is not divisible by 6 in $\mathbb{Z}$. But over $\mathbb{R}$ one can write

$$32 = 6 \cdot \frac{16}{3} = 6l^6,$$

if we set

$$l^6 = \frac{16}{3}.$$

This is not a problem for the real model because it uses equality over $\mathbb{R}$, not a divisibility statement in the ring of integers.

Therefore, the correct formulation is:

$$l^n = \frac{S_n(m,p)}{n} \quad \text{as a real definition,}$$

but not necessarily

$$n \mid S_n(m,p) \quad \text{as an integer statement.}$$

9 The Second Real Normalization: $S_n(m,p) = 2^{n-1}q^n$

Now let's look not at the multiplier n, but at the total mass of odd binomial coefficients.

From the binomial identity it is known:

$$\sum_{\substack{1\le r\le n\\ r \text{ odd}}} \binom{n}{r} = 2^{n-1}. \tag{17}$$

This means: the sum of odd coefficients in a row of Pascal's triangle is equal to half of the full sum of the coefficients.

Therefore, we can introduce a second positive real residual scale $q > 0$:

$$S_n(m,p) = 2^{n-1}q^n. \tag{18}$$

Equivalently,

$$q^n = \frac{S_n(m,p)}{2^{n-1}}. \tag{19}$$

Here q is neither a module nor a remainder modulo. It is a real scaling parameter that remains after extracting the mass of odd binomial coefficients 2^{n-1}.

From $y^n = 2S_n(m,p)$ and (18) we obtain

$$y^n = 2^n q^n. \tag{20}$$

In Coq, this normalization corresponds to the definition `coefficient_mass_factorization`. Its meaning is:

$$0 < n, \qquad \text{core} = 2^{n-1}q^n$$

over the real numbers.

10 Comparison of the Two Normalizations

Now the same core $S_n(m,p)$ is written in two ways:

$$S_n(m,p) = n\,l^n, \qquad S_n(m,p) = 2^{n-1}q^n. \tag{21}$$

Equating the right sides, we get

$$n\,l^n = 2^{n-1}q^n.$$

Since $n > 0$, we can divide by n:

$$l^n = q^n\frac{2^{n-1}}{n}. \tag{22}$$

This is not a hypothesis, but a direct consequence of the two real factorizations of the same core. This is exactly what the Coq lemma `two_scale_factorizations_relation` verifies.

Contextually, formula (22) says: two normalizations of a single core generally yield different residual scales. Their ratio is governed by the multiplier

$$\frac{2^{n-1}}{n}.$$

11 The Coefficient Symmetry Compatibility Obstacle

After comparing the two normalizations, we obtained the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}. \tag{23}$$

This relation is the bridge to the logic of version 15.3. It shows that the equality of residual scales

$$l^n = q^n \tag{24}$$

is not an isolated numerical coincidence. Under two normalizations of the same core, it is equivalent to the equality of the two extracted coefficient masses:

$$n = 2^{n-1}.$$

Let us prove this in both directions.

First direction. Let $l^n = q^n$ hold. Then from (23) we obtain

$$q^n = q^n \frac{2^{n-1}}{n}.$$

Since $q > 0$, then $q^n > 0$, and division by q^n is legitimate. We are left with

$$1 = \frac{2^{n-1}}{n},$$

that is

$$n = 2^{n-1}. \tag{25}$$

Reverse direction. If, conversely,

$$n = 2^{n-1},$$

then the multiplier

$$\frac{2^{n-1}}{n}$$

is equal to one. Then the exact relation (23) immediately turns into

$$l^n = q^n.$$

Thus, under two real normalizations of the same odd binomial core, we obtain the exact equivalence

$$l^n = q^n \iff n = 2^{n-1}. \tag{26}$$

For a lyceum student, this can be read as: the two residual scales coincide exactly when the two extracted coefficient masses coincide: the linear coefficient n and the total mass of odd binomial coefficients 2^{n-1}. For $n = 2$ they coincide. For $n > 2$, the mass 2^{n-1} becomes larger than n, and the two scales can no longer remain equal.

In Coq, the simple equality of residual scales is defined as `residual_scale_equalitynlq`. The equality of coefficient masses is defined as `coefficient_mass_equalityn`. The central theorem `coefficient_symmetry_compatibility_iff` formalizes precisely (26).

12 The Quadratic Case and the Divergence of Scales for Large Exponents

Let's consider what the equivalence

$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1}$$

means for the first values of n.

Quadratic case. For $n = 2$, the corresponding row of Pascal's triangle is

$$1, \quad 2, \quad 1.$$

In the difference of two symmetric squares, only one odd binomial coefficient remains:

$$\binom{2}{1} = 2.$$

The first normalization extracts

$$n = 2,$$

and the second normalization extracts

$$2^{2-1} = 2.$$

Both coefficient masses coincide:

$$n = 2^{n-1}.$$

Therefore, the two residual scales can coincide:

$$l^2 = q^2.$$

This is exactly the quadratic, Pythagorean case.

Cubic case. For $n = 3$, the odd binomial coefficients are

$$\binom{3}{1} = 3, \qquad \binom{3}{3} = 1.$$

Their total mass is

$$2^{3-1} = 4.$$

The first normalization extracts 3, and the second extracts 4. Therefore

$$l^3 = q^3 \frac{4}{3},$$

and, for $q > 0$,

$$l^3 \neq q^3.$$

Fourth power. For $n = 4$, the odd coefficients are

$$\binom{4}{1} = 4, \qquad \binom{4}{3} = 4.$$

Their total mass is

$$2^{4-1} = 8.$$

The first normalization extracts 4, and the second extracts 8. Therefore

$$l^4 = 2q^4,$$

and again

$$l^4 \neq q^4.$$

In general, for every integer $n > 2$ we have

$$2^{n-1} > n.$$

Consequently, from

$$l^n = q^n \frac{2^{n-1}}{n}$$

with $q > 0$ it follows that the coefficient multiplier on the right is greater than one. This is exactly what is called the coefficient symmetry compatibility obstacle: the equality $l^n = q^n$ is compatible with the two normalizations only for $n = 1$ and $n = 2$.

13 Why $n = 2^{n-1}$ Yields Only $n = 1, 2$

Consider the equation

$$n = 2^{n-1}.$$

For $n = 1$ we obtain

$$1 = 2^0 = 1.$$

For $n = 2$ we obtain

$$2 = 2^1 = 2.$$

That is, $n = 1$ and $n = 2$ are indeed solutions.

Let us show that there are no solutions for $n > 2$. Equivalently, we can write

$$2^n = 2n.$$

But for all integers $n \geq 3$ we have

$$2^n > 2n. \tag{27}$$

For example:

$$2^3 = 8 > 6, \qquad 2^4 = 16 > 8, \qquad 2^5 = 32 > 10.$$

Further on, the gap only grows because the exponential function doubles at each step, while the linear function only adds a constant value.

In Coq language, this part is split into several elementary lemmas:

- `pow2_gt_linear_shift`;
- `pow2_gt_linear`;
- `pow_eq_linear_positive`;
- `two_pow_eq_two_mul_iff_shift`;
- `binary_scaling_roots_only_one_two`.

The resulting formal thought is simple:

$$n = 2^{n-1} \quad \Longrightarrow \quad n = 1 \text{ or } n = 2.$$

14 The Conditional Theorem of the Model

Theorem 14.1 (Conditionally under compatibility of residual scales)**.** *Suppose a hypothetical positive integer solution to Fermat's equation*

$$x^n + y^n = z^n, \qquad n > 2,$$

admits a real symmetric parameterization

$$z = m^n + p^n, \qquad x = m^n - p^n.$$

Suppose the odd binomial core $S_n(m,p)$ admits two real normalizations

$$S_n(m,p) = n\,l^n, \qquad S_n(m,p) = 2^{n-1}q^n,$$

where $l > 0$, $q > 0$. Then a comparison of the two normalizations yields

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Consequently,

$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1}.$$

Since the equation $n = 2^{n-1}$ has only the solutions $n = 1$ and $n = 2$ among positive integers, the equality of residual scales is incompatible with any exponent $n > 2$.

Proof. From the two normalizations it follows

$$n\,l^n = 2^{n-1}q^n,$$

which means

$$l^n = q^n \frac{2^{n-1}}{n}.$$

If $l^n = q^n$, then

$$q^n = q^n \frac{2^{n-1}}{n}.$$

Since $q > 0$, then $q^n > 0$, and division by q^n is legitimate. We obtain

$$n = 2^{n-1}.$$

Conversely, if $n = 2^{n-1}$, then the exact relation immediately yields $l^n = q^n$. Therefore, the equality of residual scales is equivalent to the equation $n = 2^{n-1}$. But this equation has only the solutions $n = 1$ and $n = 2$ in positive integers. Thus, the case $n > 2$ contradicts the requirement of compatibility of residual scales. □

This theorem is conditional. The main mathematical question is whether the internal bi-power symmetry of a hypothetical positive integer solution is obliged to preserve the equality $l^n = q^n$ between two natural normalizations of the same odd binomial core.

15 The Modular Caveat (Remark)

In discussions of FLT, people often cite examples of congruences of the form

$$a^n + b^n \equiv c^n \pmod{\text{modq}}.$$

Such examples can be correct, but they belong to a different level of the problem.

A modular congruence means there exists an integer t such that

$$c^n = a^n + b^n + \text{modq} \cdot t.$$

An equality in integers is only a special case where $t = 0$.

Example 15.1. We have

$$1^3 + 1^3 = 2, \qquad 3^3 = 27.$$

In integers $2 \neq 27$. But modulo 5

$$27 = 2 + 5 \cdot 5,$$

meaning

$$3^3 \equiv 1^3 + 1^3 \pmod{5}.$$

This is a true modular congruence, but not a true integer equality.

Therefore, the existence of solutions to congruences

$$x^n + y^n \equiv z^n \pmod{\text{modq}}$$

for $n > 2$ is not a counterexample to Fermat's Last Theorem and does not refute the real parameterization.

If one deliberately moves to the modular level, the parameters must be specified separately:

$$m^n \equiv A \pmod{\text{modq}}, \qquad p^n \equiv B \pmod{\text{modq}},$$

and only then write

$$z \equiv A + B \pmod{\text{modq}}, \qquad x \equiv A - B \pmod{\text{modq}}.$$

16 How the Modular Caveat is Reflected in Coq

In Coq, there is a section `ModularRemark`. In it, the relation of modular congruence is defined as

`modular_congruentmodqab`.

Mathematically this means:

$$\exists t \in \mathbb{Z} : \quad a = b + \text{modq} \cdot t.$$

The lemma `integer_equality_implies_modular_power_congruence` verifies the direction:

$$\text{integer equality} \quad \Longrightarrow \quad \text{modular congruence}.$$

If

$$x^n + y^n = z^n,$$

then one can take $t = 0$.

The lemma `integer_equality_is_zero_modular_witness` explicitly records that a true equality corresponds to a zero modular witness $t = 0$.

The lemma `modular_congruence_not_integer_equality` verifies the example:

$$3^3 \equiv 1^3 + 1^3 \pmod 5,$$

but

$$3^3 \neq 1^3 + 1^3$$

in integers.

The structure `ModularResidues` formalizes the idea that at the modular level one must separately set the residues of m^n and p^n, and then separately set the residues of z and x.

17 Connection to Key Coq Fragments

This section lists the main Coq objects of version 15.3 and their meaning.

`sum_diff_from_parameters_R`
Verifies $z + x = 2m^n$ and $z - x = 2p^n$ over $\mathbb{R}$.

`sum_diff_from_parameters_Z`
The same over $\mathbb{Z}$.

`parity_condition_Z`
If $m, p \in \mathbb{Z}$, then $z + x$ and $z - x$ must be even.

`no_parameters_if_parity_violation`
If the parity of $z + x$ or $z - x$ is violated, there are no integer parameters m, p in this form.

`no_parameters_if_odd`
The same idea written through the oddness of the corresponding sums and differences.

`no_parameters_for_example`
A formal example $z = 2$, $x = 1$, $n = 3$: real reconstruction is possible, but integer is impossible.

`real_core_factorization`
First normalization core $= n\, l^n$.

`coefficient_mass_factorization`
Second normalization core $= 2^{n-1} q^n$.

`two_scale_factorizations_relation`
Derives the exact relation $l^n = q^n \frac{2^{n-1}}{n}$ from two factorizations of one core.

`residual_scale_equality`
Equality of residual scales $l^n = q^n$.

`coefficient_mass_equality`
Equality of coefficient masses $n = 2^{n-1}$.

`residual_scale_equality_forces_coefficient_mass_equality`
Derives $n = 2^{n-1}$ from two factorizations, the equality $l^n = q^n$, and positivity $q > 0$.

`coefficient_mass_equality_forces_residual_scale_equality`
Derives $l^n = q^n$ from two factorizations and the equality $n = 2^{n-1}$.

`coefficient_symmetry_compatibility_iff`
Verifies the central equivalence of version 15.3: $l^n = q^n \iff n = 2^{n-1}$.

`CoefficientSymmetryData`
Packages n, core, l, q, positivity, and compatibility conditions of the model.

`coefficient_symmetry_data_forces_shift`
Derives $n = 2^{n-1}$ from `CoefficientSymmetryData`.

`binary_scaling_roots_only_one_two`
Derives $n = 1$ or $n = 2$ from $n = 2^{n-1}$.

`coefficient_symmetry_data_forces_small_exponent`
Derives $n = 1$ or $n = 2$ from compatibility data.

`coefficient_symmetry_data_excludes_high_exponent`
Derives a contradiction from compatibility data and $n > 2$.

`coefficient_symmetry_compatibility_excludes_high_exponent`
Formalizes the same final prohibition directly from two factorizations, $l^n = q^n$, $q > 0$, and $n > 2$.

18 What Exactly Coq Verifies

Coq does not verify the entire historical hypothesis nor the unconditional FLT. It verifies the formal conditional chain of version 15.3.

First, two real factorizations

$$\text{core} = n\, l^n, \qquad \text{core} = 2^{n-1} q^n$$

entail the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Further, Coq verifies the compatibility equivalence

$$l^n = q^n \quad \iff \quad n = 2^{n-1}.$$

In terms of Coq, this is written as

$$\texttt{residual\_scale\_equalitynlq} \iff \texttt{coefficient\_mass\_equalityn}.$$

Then Coq verifies the elementary arithmetic part:

$$n = 2^{n-1} \quad \Longrightarrow \quad n \in \{1, 2\}.$$

Consequently,

$$n > 2$$

is impossible under two factorizations, positivity $q > 0$, and the requirement $l^n = q^n$.

In other words, Coq confirms the absence of logical errors in the final conditional conclusion. But the substantive reason why a hypothetical positive integer solution must preserve the equality $l^n = q^n$ needs to be discussed separately.

19 What Coq Does Not Prove

To be fair, it must be explicitly stated: Coq does not automatically prove that from every hypothetical solution of the equation

$$x^n + y^n = z^n$$

the necessity to preserve the equality of residual scales

$$l^n = q^n$$

between two natural real normalizations of the same odd binomial core follows.

Coq verifies something else: if two factorizations are accepted

$$\text{core} = n\, l^n, \qquad \text{core} = 2^{n-1} q^n,$$

if it is accepted that

$$l^n = q^n,$$

and if $q > 0$, then the further conclusion is strict:

$$n = 2^{n-1} \quad \Longrightarrow \quad n \in \{1, 2\}.$$

Therefore, the weak point of the conditional scheme is not in the arithmetic of the equation $n = 2^{n-1}$ and not in the Coq-verifiable final chain. The decisive point remains the mathematical status of the requirement: must the internal bi-power symmetry of a hypothetical positive integer solution preserve the equality $l^n = q^n$?

20 Why Compilation Time Is Important but Does Not Replace Mathematics

If a Coq file compiles in a few seconds, it means that the formally written lemmas and theorems have been verified by the machine. This is a good technical sign: there are no missing proofs, illegal transitions, or unclosed goals in the code.

However, compilation time is not a proof of the original mathematical hypothesis. The machine checks only what is written in the formal language. If the equality of residual scales $l^n = q^n$ is introduced as a compatibility requirement, Coq honestly verifies the consequences of this requirement, but does not derive it out of thin air.

Therefore, the correct formula is:

> A quick Coq compilation confirms the correctness of the formal deduction within the specified prerequisites, but does not turn a conditional prerequisite into an unconditional theorem.

21 How to Properly Phrase a Message for a Forum

For a mathematical forum, it is better to use a calm and precise phrasing:

> I propose discussing a conditional reconstructive model surrounding FLT. In it, the odd binomial core $S_n(m, p)$ is written in two ways: $S_n(m, p) = n\, l^n$ and $S_n(m, p) = 2^{n-1} q^n$. Comparing these normalizations yields $l^n = q^n \frac{2^{n-1}}{n}$. Therefore, under two normalizations of the same core, the equality of residual scales $l^n = q^n$ is equivalent to $n = 2^{n-1}$. This equation has only

> positive solutions $n = 1$ and $n = 2$. Coq verifies the final formal chain, but the main question remains mathematical: is the internal bi-power symmetry of a hypothetical positive integer solution obliged to preserve the equality of residual scales?

This style is honest and constructive. It does not demand that the reader immediately accept the entire model, but invites them to check a specific central point.

22 Frequently Asked Questions (FAQ)

22.1 Does this educational file unconditionally prove FLT?

No. This is an educational explanation of a conditional model. The unconditional FLT has been proven by Wiles and Taylor. A different conditional scheme is considered here.

22.2 What is the main mathematical node of version 15.3?

The main node is the status of the equality of residual scales

$$l^n = q^n.$$

Version 15.3 shows that under two natural real normalizations of the same odd binomial core, this equality is equivalent to

$$n = 2^{n-1}.$$

Meaning, it is compatible only with $n = 1$ and $n = 2$. Open for discussion remains the question of whether a hypothetical positive integer solution is obliged to preserve this equality of residual scales.

22.3 What exactly does Coq verify?

Coq verifies the formal chain:

$$\text{core} = n\, l^n, \qquad \text{core} = 2^{n-1} q^n \quad \Longrightarrow \quad l^n = q^n \frac{2^{n-1}}{n},$$

and then the central equivalence

$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1},$$

and the final conclusion

$$n = 2^{n-1} \quad \Longrightarrow \quad n \in \{1, 2\}.$$

22.4 Why is q not a module?

In this model, q is the real residual scale of the second normalization. The module in modular arithmetic is denoted by modq.

22.5 Why is real factorization not divisibility?

Because
$$S_n(m,p) = n\, l^n$$
over $\mathbb{R}$ means only a real equality. It does not automatically mean
$$n \mid S_n(m,p)$$
in $\mathbb{Z}$.

22.6 Why can't the equality $l^n = q^n$ just be declared obvious?

Because l and q are residual scales after two different normalizations of the same core. Their equality is a strong compatibility requirement. Version 15.3 clarifies its exact meaning:
$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1}.$$
This equivalence explains why the equality holds in the quadratic case and inevitably fails for higher exponents.

22.7 Why is the quadratic case special?

For $n = 2$, the linear coefficient is 2, and the total mass of odd binomial coefficients is also $2^{2-1} = 2$. Therefore, the two extracted coefficient masses coincide, and the residual scales can coincide:
$$l^2 = q^2.$$
For $n > 2$, the total mass of odd coefficients 2^{n-1} is strictly greater than n, so the two residual scales diverge.

23 Educational Exercises

1. Expand $(A+B)^5 - (A-B)^5$ and verify that only odd powers of B remain.
2. For $n = 4$, write out the odd core $S_4(m,p)$ explicitly.
3. Prove the identity
$$\sum_{\substack{1 \le r \le n \\ r \text{ odd}}} \binom{n}{r} = 2^{n-1}$$
using the equalities $(1+1)^n$ and $(1-1)^n$.
4. For $n = 6$, $m = 1$, $p = 1$, verify that the odd sum of coefficients equals 32, and explain why this is not divisible by 6 in $\mathbb{Z}$.
5. Show by induction that for all $n \ge 3$, it holds that $2^n > 2n$.
6. Explain in your own words how an equality over $\mathbb{Z}$ differs from a modular congruence.
7. Using the formula
$$l^n = q^n \frac{2^{n-1}}{n},$$
explain why $l^n = q^n$ is equivalent to $n = 2^{n-1}$.
8. Check separately the cases $n = 2$, $n = 3$, and $n = 4$, and explain why the quadratic case is special.

24 Summary Diagram of the Entire Model

In a compressed form, the entire conditional scheme looks like this:

$$x^n + y^n = z^n$$

$$\Downarrow$$

$$y^n = (m^n + p^n)^n - (m^n - p^n)^n$$

$$\Downarrow$$

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m,p)$$

$$\Downarrow$$

$$S_n(m,p) = n\,l^n, \qquad S_n(m,p) = 2^{n-1}q^n$$

$$\Downarrow$$

$$l^n = q^n \frac{2^{n-1}}{n}$$

$$\Downarrow$$

$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1}$$

$$\Downarrow$$

$$n \in \{1,2\}$$

$$\Downarrow$$

for $n > 2$ the equality of the residual scales
is incompatible with the two normalizations.

25 Conclusion

Version 15.3 of the educational file explains the scheme of the coefficient symmetry compatibility obstacle. The main logic consists in comparing two natural real normalizations of the same odd binomial core:

$$S_n(m,p) = n\,l^n, \qquad S_n(m,p) = 2^{n-1}q^n.$$

From them follows the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Therefore, in version 15.3, the equality of residual scales has a precise meaning:

$$l^n = q^n \quad \Longleftrightarrow \quad n = 2^{n-1}.$$

And this is possible only for $n = 1$ and $n = 2$.

Coq verifies the final formal chain, but does not replace the proof that a hypothetical positive integer solution must preserve the equality $l^n = q^n$. Therefore, the correct status of the model is a conditional reconstruction: under the requirement of compatibility of residual scales, an exponent $n > 2$ is impossible.

26 Mini-Glossary

Term	**Meaning**
Fermat's Last Theorem	The statement that $x^n + y^n = z^n$ has no positive integer solutions for $n > 2$.
Binomial theorem	The expansion formula for $(A + B)^n$ via binomial coefficients.
Odd binomial core	The sum of odd terms remaining in $(m^n + p^n)^n - (m^n - p^n)^n$ after canceling the even terms.
First normalization	The notation $S_n(m, p) = n\, l^n$, where the linear binomial coefficient n is isolated.
Second normalization	The notation $S_n(m, p) = 2^{n-1}q^n$, where the total mass of odd binomial coefficients is isolated.
Real factorization	A notation over $\mathbb{R}$; not the same as divisibility in $\mathbb{Z}$.
Residual scale	A positive real scale remaining after extracting the coefficient mass from the same core.
Equality of residual scales	The equality $l^n = q^n$.
Coefficient symmetry compatibility obstacle	A fact of version 15.3: given two normalizations of one core, $l^n = q^n \iff n = 2^{n-1}$.
Module modq	The number modulo which residues are considered; it is not the scale q.
`coefficient_symmetry_compatibility_iff`	Coq theorem formalizing the equivalence between the equality of residual scales and the equality of coefficient masses.
`CoefficientSymmetryData`	Coq structure combining n, the core, scales l, q, positivity, the two factorizations, and the equality of residual scales.
Coq	An interactive theorem proving system that verifies formal proofs.

27 List of References and Guidelines

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